

Statistical Modelling and Inference for Health - Assessment 2

Student ID: 14210037

Introduction

A critical disease, heffpox, has recently emerged and is causing a high risk of early death among affected patients. A new treatment, Milnepan, has been introduced and is thought to reduce this short-term mortality risk, but its effectiveness in routine clinical practice remains uncertain. The aim of this study is to evaluate whether prescribing Milnepan reduces the 7-day mortality among patients with heffpox after adjusting for baseline confounders at Hundred Acre Hospital.

Methods and Data Description

I. Data and Variables

Data represented patients from Hundred Acre Hospital during the heffpox outbreak. Baseline characteristics included sex, age, BMI, smoking status, diabetes; treatment variables were Milnepan prescription and intensive care unit (ICU) admission. Exploratory analysis of treatment group differences justifies confounder adjustment. TEVENT (admission to death/censoring) suitable for survival analysis; death7day appropriate for causal inference on binary outcome.

II. Missing Data Strategy

Missing data handling will rely on exploratory analysis patterns, considering the missing rate and machine. If missingness is plausibly missing completely at random (MCAR), complete-case analysis will be used and expected to yield unbiased estimates. If missingness is consistent with missing at random (MAR), where missingness depends on observed data, multiple imputation by chained equations (MICE) will be applied to generate imputed datasets via stochastic regression, validating by conditional distributions of missing variables given observed data. Moreover, estimates will be combined using Rubin's rules. Complete-case analyses will also be conducted as a sensitivity analysis, and potential residual bias under missing not at random (MNAR) mechanisms will be acknowledged but not explicitly modelled.

III. Statistical methods

For the survival analysis, time to event was summarised using Kaplan-Meier curves and compared between treatment groups using the log-rank test, following by non-informative censoring assumption. For the causal analysis of death within 7 days, we estimated the effect of Milnepan using three approaches: multivariable logistic regression, propensity score (PS) matching, and inverse probability of treatment weighting (IPTW).

For multivariable logistic regression, this study adjusted for all baseline

confounders in a single model. This approach assumes correct model specification and that the relationship between covariates and outcome follows a logistic form. Propensity score (PS) matching and Inverse Probability of Treatment Weighting (IPTW) rely on four key assumptions for causal inference: (1) no unmeasured confounding because all variables affecting both treatment and outcome have been measured and adjusted for; (2) every patient has a non-zero probability of receiving either treatment or control; (3) the treatment is well-defined and potential outcomes are consistent with observed outcomes under each treatment condition; and (4) after conditioning on measured confounders, treatment groups are comparable, indicating exchangeability. For PS matching, first estimated each patient's propensity score using logistic regression, then performed 1:1 nearest-neighbour matching without replacement. Covariate balance was assessed using standardized mean differences (SMD), with SMD closed to 0 considered well balance. For IPTW, this analyzes calculate stabilized inverse probability weights using the estimated propensity scores. Each patient was weighted by the inverse of their probability of receiving their actual treatment, with weights stabilized by the marginal probability of treatment to reduce variance. The study assessed weight distributions to identify potential violations of the positivity assumption and evaluated covariate balance using weighted SMDs.

Results

I. Descriptive Analysis

Table 1. Baseline characteristics and outcomes by treatment (milnepan)

Variable	Category	Overall	No milnepan	Milnepan
Sample size		8192 (100.0%)	4244 (51.8%)	3948 (48.2%)
Demographics and Baseline Characteristics				
Age (years)*		43.0 (30.0–55.0)	46.0 (33.0–59.0)	39.0 (28.0–51.0)
BMI (kg/m ²)*		28.0 (25.3–30.7)	27.9 (25.2–30.5)	28.1 (25.4–30.8)
Time-to-event (days)*		10.0 (3.1–10.0)	8.5 (2.5–10.0)	10.0 (3.9–10.0)
Sex	Female	4207 (51.4%)	2243 (52.9%)	1964 (49.7%)
	Male	3985 (48.6%)	2001 (47.1%)	1984 (50.3%)
Smoking status	No current smoker	5896 (72.0%)	3000 (70.7%)	2896 (73.4%)
	Smoker	1913 (23.4%)	1050 (24.7%)	863 (21.9%)
	Missing	383 (4.7%)	194 (4.6%)	189 (4.8%)
Diabetes	No	7340 (89.6%)	3617 (85.2%)	3723 (94.3%)
	Yes	852 (10.4%)	627 (14.8%)	225 (5.7%)
Clinical Outcomes				
ICU admission	No	7379 (90.1%)	3764 (88.7%)	3615 (91.6%)
	Yes	813 (9.9%)	480 (11.3%)	333 (8.4%)
Vital status	Censored	4105 (50.1%)	1922 (45.3%)	2183 (55.3%)
	Died	4087 (49.9%)	2322 (54.7%)	1765 (44.7%)
Death within 7 days	Alive ≥7 days	4886 (59.6%)	2347 (55.3%)	2539 (64.3%)
	Death <7 days	3306 (40.4%)	1897 (44.7%)	1409 (35.7%)

Note:

BMI missing in 2686 patients (32.8%); smoking status missing in 383 patients (4.7%).

For continuous variables* show median (IQR); categorical variables indicate number (percentage).

Table 1 and Figure 1 illustrate the distribution of the data, including overall and separated by treatment. This dataset includes 8192 records, with 3948 patients (48.2%) in the treatment group. The distribution of age was right-skewed, obviously, a median age of 43 years old (IQR 30 - 55 years old). The population who took the medicine was younger (median = 39 years old, IQR 28-51

years old) than the control group (median = 46 years old, IQR 33-59 years old). Although BMI seemed to follow a normal distribution, and medians were similar, the records had missed 2686 (32.8%). The days patients died after admission in the

control group were fewer (8.5 days, IQR 2.5 - 10 days) compared with the treatment group (10 days, IQR 3.9 - 10 days). However, 4105 people (50.11%) were recorded 10 days, which included censoring. The gender distribution was approximately balanced, with differences of about 80 females and about 20 males between the treatment and control groups. 72.0 % of the sample population were not current smokers, with 4.7% of the missing on smoking status. Diabetes prevalence was 10.4% overall and was approximately 2.6 times higher in the control group (14.8%) than in the treatment group (5.7%). Only 9.9% of the population had ICU experience, with a slightly higher proportion in the control group (11.3%). The horrible disease caused 49.9% population to die based on study records, with 54.7% in the non-treatment group. The primary outcome of death within 7 days was recorded 40.4%, with 44.7% in the control group and 35.7% in the intervention group.

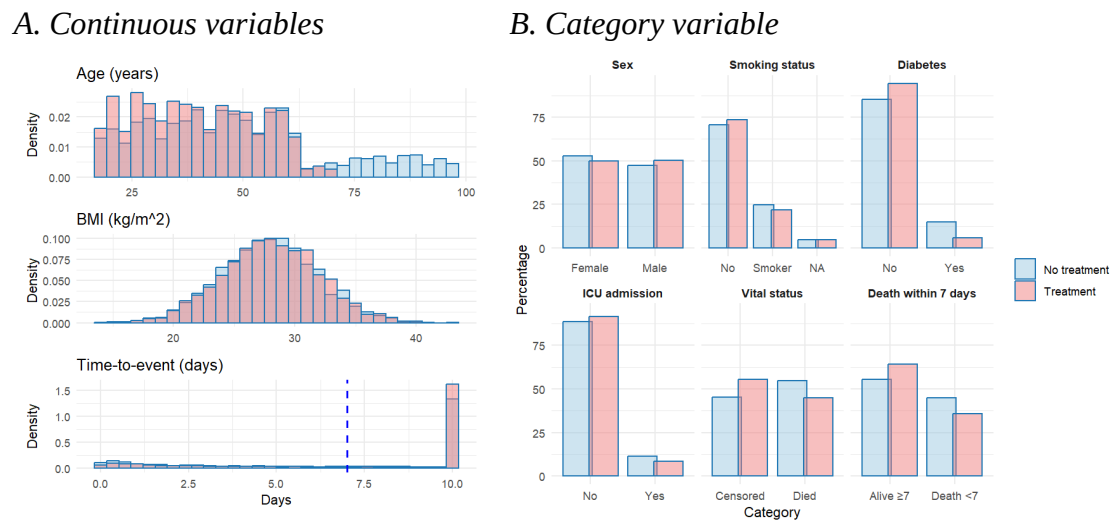


Figure 1. The distribution of variables.

II. Survival Analysis by log-rank test and Kaplan-Meier curves for TEVENT

Table 2. Log-rank test comparing survival by milnepan status

Group	N	Observed deaths	Expected deaths	Chi-square	df	p value
No milnepan	4302	2322	1846.098	224.1	1	0.000
Milnepan	4587	1765	2240.902	-	-	-

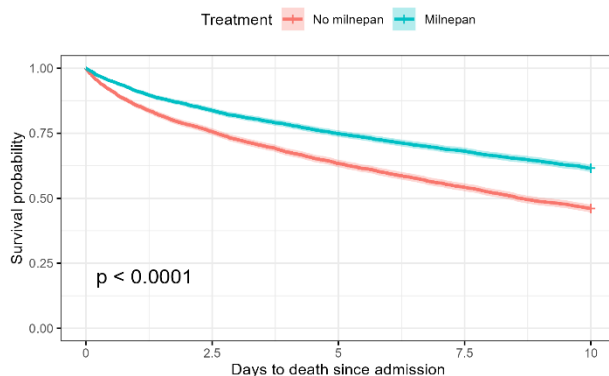


Figure 2. Kaplan-Meier curves of TEVENT by milnepan treatment status

Since survival analysis focuses on treatment status, time-to-event, and vital status, Kaplan-Meier (KM) curves and log-rank tests were performed using the original dataset. The result of log-rank test showing on Table 2 indicated a highly statistically significant difference in survival between patients treated with Milnepan and those not treated (log-rank $\chi^2 = 224$, $df = 1$, $p < 2 \times 10^{-16}$). In the control group, 2322 deaths were observed compared

with 1846 expected under the null hypothesis of equal survival, whereas in the Milnepan group, 1765 deaths were observed compared with 2241 expected. These patterns, together with the progressively diverging KM curves in Figure 2, indicate substantially better survival among patients receiving Milnepan compared with those not treated.

III. Imputed Missing Data

Based on the missing data pattern, 5280 participants (64.5%) had complete data, 2529 (30.9%) were missing BMI only, 226 (2.8%) were missing smoking status only, and 157 (1.9%) were missing both variables.

Table 3. Finding the association for the missing data and patient characteristics

A. Associations between BMI missingness and patient characteristics

Variable	Test	Test statistic	df	p-value	Signif.
Age (years)	t-test	-20.02	4702	<0.001	***
Time-to-event (days)	t-test	3.23	5212	0.001	**
Sex	Chi-squared	2.86	1	0.091	
Diabetes	Chi-squared	269.31	1	<0.001	***
Milnepan use	Chi-squared	14.19	1	<0.001	***
ICU admission	Chi-squared	2.22	1	0.136	
Vital status	Chi-squared	12.28	1	<0.001	***
Death within 7 days	Chi-squared	3.60	1	0.058	
Smoking status	Chi-squared	6.68	1	0.010	**

B. Associations between smoking status missingness and patient characteristics

Variable	Test	Test statistic	df	p-value	Signif.
Age (years)	t-test	-5.93	416	<0.001	***
Time-to-event (days)	t-test	0.63	421	0.530	
BMI (kg/m ²)	t-test	-0.31	242	0.759	
Sex	Chi-squared	1.37	1	0.241	
Diabetes	Chi-squared	14.66	1	<0.001	***
Milnepan use	Chi-squared	0.17	1	0.681	
ICU admission	Chi-squared	0.01	1	0.929	
Vital status	Chi-squared	1.97	1	0.160	
Death within 7 days	Chi-squared	0.91	1	0.341	

Based on Table 3, the study explored the association between missingness and population features. Missing BMI values were more frequent among older patients (mean age 50.1 vs 41.5 years), among those without diabetes (2.5% vs 14.3%), among those not receiving Milnepan (54.8% vs 50.3%), and among those who died during follow-up (41.8% vs 39.6%), and were also associated with a shorter time to event (mean 6.68 vs 6.97 days) and with current smoking (26.3% vs 23.6%). Patients with missing smoking status were on average older than those with observed smoking status (49.8 vs 44.1 years), and missing smoking information was less common among patients with diabetes than among those without diabetes (4.4% vs 10.7%). Taken together, these patterns indicate that both BMI and smoking status are compatible with a MAR mechanism conditional on the observed covariates. Under this assumption, multiple imputation is an appropriate strategy to handle missingness in these variables.

Multiple imputation with 20 datasets was conducted using the mice package in R. While predictive mean matching (pmm) was used for BMI, treated as a continuous variable, and logistic regression (logreg) was used for smoking, treated as a binary variable. All available variables were included in the imputation model to improve the efficiency of the imputations. For the confirmatory dataset, 20 datasets was conducted

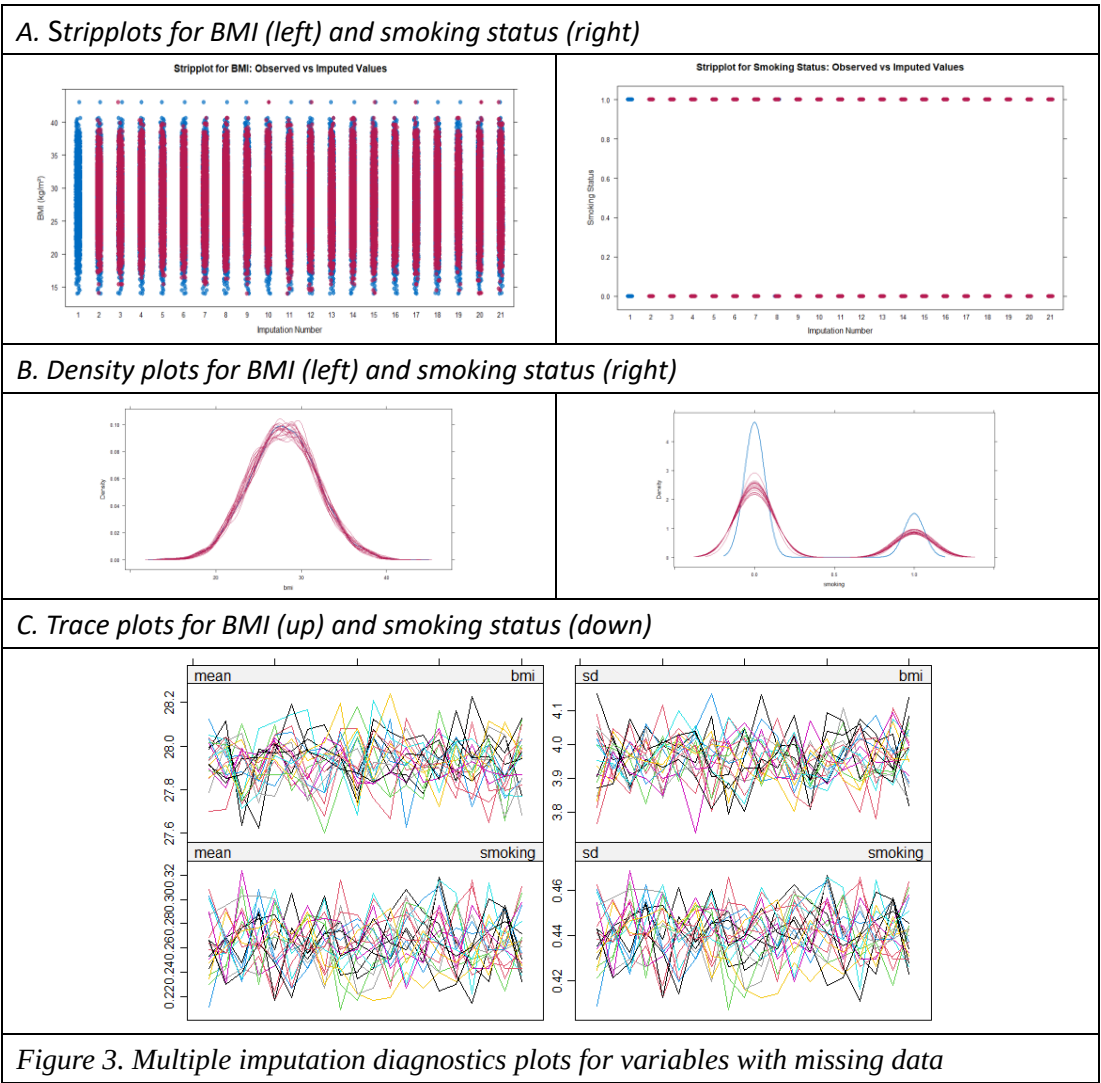
to ensure adequate stability of pooled estimates.

To assess the quality of the imputed datasets, Figure 3 showed diagnostic plots generated by the mice package. Stripplot showed that imputed values fell within the range of observed values and followed a similar distribution across treatment groups. Density plots confirmed that the distributions of imputed values closely matched those of observed values for both variables. Trace plots demonstrated stable convergence of the imputation algorithm across iterations, with no systematic trends or divergence in means or standard deviations. These diagnostics support the validity of the imputation model and suggest that the imputed values are plausible given the observed data.

Table 4. Original vs Imputed Logistic Regression for Milnepan

Model	OR for milnepan	95% CI	p-value
Original data	0.88	0.88 (0.79-0.99)	0.040
Imputed (pooled)	0.87	0.87 (0.79-0.96)	0.004

Because the 7 days mortality is binary data the logistic was been conducted to analysis the quality of imputed data. The result showed that original OR was 0.88 (95%CI 0.79 - 0.99) comparing with pooled imputed data was 0.87 (95%CI 0.79 - 0.96), statistically significant. Thus, the imputed data will be used to evaluate causality.



IV. Causal Analysis

(1) Causal framework

This study assesses Milnepan's causal effect on 7-day mortality in heffpox patients. Figure 4 presented the directed acyclic graph (DAG) motivated by clinical question. In the DAG, Milnepan was the treatment and death within 7 days is the outcome. ICU admission is treated as a mediator because it typically occurs after treatment initiation and may lie on the causal pathway between Milnepan and early death. Baseline characteristics such as age, sex, BMI, smoking status, and diabetes were assumed to influence both treatment decisions and the risk of death, and were therefore considered confounders to be adjusted for in the causal analyses.

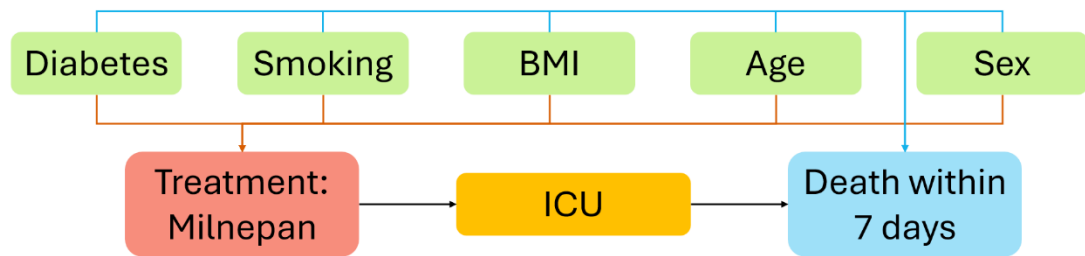


Figure 4. Directed acyclic graph (DAG) for Milnepan to Death within 7 days
Green: confounders, Red: treatment, Orange: Mediator, Blue: outcome

Prior to causal inference, key assumptions were verified for valid causal estimation. Examination of propensity score distributions confirmed adequate overlap between treatment groups across all covariate levels, supporting the positivity assumption. The directed acyclic graph (Figure 4) identified age, sex, BMI, smoking status, and diabetes as the minimal sufficient set of confounders, and treatment was well-defined per hospital protocols, satisfying the consistency assumption.

(1) Multi-imputed data with multivariable logistic regression

$$\begin{aligned} \text{logit}\{P(\text{death7day} = 1 \mid \text{milnepan}, C)\} \\ = \beta_0 + \beta_1 \text{milnepan} + \beta_2 \text{age} + \beta_3 \text{sex} + \beta_4 \text{bmi} + \beta_5 \text{smoking} \\ + \beta_6 \text{diabetes}, \quad \text{where } C = (\text{age, sex, bmi, smoking, diabetes}) \end{aligned}$$

Table 5. Adjusted OR for 7-day mortality from logistic regression with MI

Variable	estimate	STD	Adjusted OR	95% CI	p value
Intercept	-3.305	0.249	0.037	0.02–0.06	0.000***
Milnepan (vs no milnepan)	-0.143	0.049	0.867	0.79–0.95	0.003**
Age (per 1 year)	0.024	0.001	1.024	1.02–1.03	0.000***
Male (vs female)	0.012	0.047	1.012	0.92–1.11	0.792
BMI (per 1 kg/m ²)	0.065	0.008	1.067	1.05–1.08	0.000***
Current smoker (vs no)	0.043	0.056	1.044	0.94–1.17	0.443
Diabetes (yes vs no)	0.539	0.076	1.715	1.48–1.99	0.000***

STD = standard error.

The adjusted logistic regression results indicated that Milnepan used was associated with an approximately 13% lower odds of death within 7 days compared with no treatment (OR 0.87, 95% CI 0.79-0.95, $p = 0.004$). Increasing age, higher BMI and pre-existing diabetes were

also independently associated with higher odds of 7-day mortality. The odds increased by about 2% per additional year of age (OR 1.02, 95% CI 1.02-1.03, $p < 0.001$), 7% per 1 kg/m² increase in BMI (OR 1.07, 95% CI 1.05-1.08, $p < 0.001$), and 71.5% for patients with diabetes compared with those without (OR 1.71, 95% CI 1.48-1.99, $p < 0.001$). In contrast, sex (OR 1.02, 95% CI 0.92-1.11, $p = 0.792$) and current smoking status (OR 1.04, 95% CI 0.94-1.17, $p = 0.443$) were not statistically significantly associated with 7-day mortality after adjustment for other covariates.

(2) Multi-imputed data with propensity score matching

Tables 6 and Figure 5 illustrated covariate balance before and after nearest-neighbour propensity score matching. Standardized mean differences improved for age (from -0.65 to 0.02) and diabetes (from -0.39 to -0.01), though moderate residual imbalance persisted. Other covariates (sex, BMI, smoking) achieved good balance with standardized mean differences between 0.06 to -0.02. Matching retained all 3,948 treated patients and 3,948 controls from the original 4,244 controls, with 296 controls discarded.

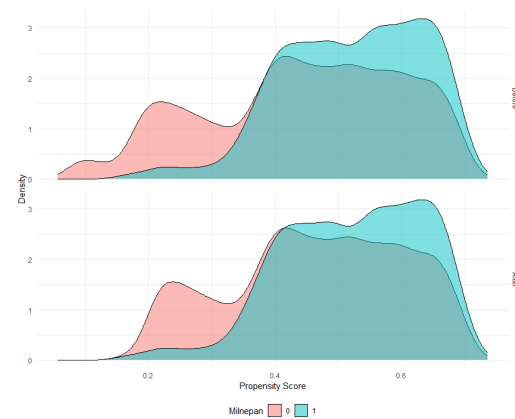


Figure 5. Pooled Distribution of Estimated Propensity Scores Before and After Matching

Table 6. Propensity Score Matching Balance Diagnostics for a Single Imputation

Variable	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio	eCDF Mean	eCDF Max
Summary of Balance for All Data						
distance	0.5223	0.4444	0.6999	0.5169	0.1409	0.2209
age	39.8121	48.5443	-0.6491	0.4481	0.1092	0.1657
sex	0.5025	0.4715	0.0621	-	0.0310	0.0310
bmi	28.1306	27.8654	0.0652	1.0581	0.0214	0.0412
smoking	0.2297	0.2611	-0.0745	-	0.0313	0.0313
diabetes	0.0570	0.1477	-0.3914	-	0.0907	0.0907
Summary of Balance for Matched Data						
distance	0.5223	0.5223	0.0000	0.9996	0.0002	0.0023
age	39.8121	39.5636	0.0185	0.8458	0.0141	0.0263
sex	0.5025	0.4721	0.0608	-	0.0304	0.0304
bmi	28.1306	27.9966	0.0330	1.0574	0.0125	0.0355
smoking	0.2297	0.2361	-0.0151	-	0.0063	0.0063
diabetes	0.0570	0.0585	-0.0066	-	0.0015	0.0015

In the result of propensity score matching, logistic regression demonstrated that Milnepan reduced 7-day mortality odds by 28% among patients with comparable baseline profiles (OR = 0.72, 95% CI: 0.66-0.79; $p < 0.001$).

(3) Multi-imputed data with Inverse Probability of Treatment Weighting (IPTW)

Table 7. Summary Statistics of IPTW Weights by Stabilization Status

Method	Min	Q1	Median	Mean	Q3	Max
Stabilized IPTW	0.55	0.78	0.93	0.99	1.15	3.51
Unstabilized IPTW	1.06	1.59	1.86	1.99	2.30	7.28

IPTW created the pseudo-population, which was not correlated with confounders, by weighting.

Table 8. Covariate balance before and after inverse probability of treatment weighting (single imputed dataset)

Covariate	Before IPTW	After IPTW
Age*	-0.511	0.062
BMI*	-0.072	0.002
Sex	Before IPTW (No/Milnepan)	After IPTW (No/Milnepan)
Male	2243/ 1964	2193.3/ 1973.9
Female	2001/ 1984	2093.0/ 1889.1
Smoking	Before IPTW (No/Milnepan)	After IPTW (No/Milnepan)
No	3145/ 3042	3242.6/ 2941.1
Yes	1099/ 906	1043.7/ 921.9
Diabetes	Before IPTW (No/Milnepan)	After IPTW (No/Milnepan)
No	3617/ 3723	3845.0/ 3474.8
Yes	627/ 225	441.3/ 388.2

Note:

Continuous variables* are summarized using absolute standardized mean differences (SMD). Categorical variables are presented as weighted counts in the IPTW pseudo-population based on stabilized weights.

The unstabilized IPTW ranged from 1.06 to 7.26 (median 1.86), and the stabilized weights from 0.55 to 3.51 (median 0.93), with a mean close to 1, suggesting no extreme weights or serious positivity violations. Table 8 was the standardized mean difference (SMD) on continuous variables and weight counts on category variables for stabilized methods only extract one imputed datasets to diagnosis the balance. Age and BMI more closed to 0, and counts were reduced the gap after IPTW. Thus indicated the imputed data after IPTW were balanced.

The fitted marginal structural model (MSM) used inverse probability of treatment weights, by regressing 7-day mortality on Milnepan in the weighted pseudo-population, with geeglm with binomial family and stabilized weights. The equation is:

$$\log\left(\frac{P_{IPTW}(death7day = 1 | milnepan)}{1 - P_{IPTW}(death7day = 1 | milnepan)}\right) = \beta_0 + \beta_1 milnepan$$

Table 9. Marginal Structural Model (MSM) with IPTW across 20 multiply imputed datasets

Variable	Estimate	STD	Wald	p-value	OR	95% CI
(Intercept)	-0.315	0.032	-9.78	0.000	0.73	0.69 – 0.78
milnepan	-0.131	0.047	-2.76	0.006	0.88	0.8 – 0.96

Table 9 presented MSM result that the IPTW odds ratio for 7-day mortality comparing Milnepan versus no treatment was 0.88 (95% CI 0.80-0.96, p = 0.007), corresponding to an approximate 12% reduction in the odds of death at 7 days

after standardizing for measured confounders via the propensity score.

(4) Complete-case sensitivity analysis

Table 10. Comparison of Causal Effect Estimates Across Different Methods

Method	Log OR	Std. Error	OR (95% CI)	p-value
Multiple Imputation Methods				
MI + Multivariable Logistic	-0.143	0.049	0.87 (0.79, 0.95)	0.0033**
MI + PSM	-0.195	0.052	0.82 (0.74, 0.91)	0.0002***
MI + IPTW	-0.131	0.048	0.88 (0.80, 0.96)	0.0059**
Complete Case Methods				
CC + Logistic	-0.124	0.060	0.88 (0.79, 0.99)	0.0398*
CC + PSM	-0.324	0.057	0.72 (0.65, 0.81)	0.0000***
CC + IPTW	-0.082	0.059	0.92 (0.82, 1.03)	0.1627

Note:

MI = Multiple Imputation (20 datasets pooled using Rubin's rules).

PSM = Propensity Score Matching (1:1 nearest neighbor).

IPTW = Inverse Probability of Treatment Weighting.

All models adjust for age, sex, BMI, smoking, and diabetes.

Table 10 illustrated using both imputed and original but merely complete-case to analyze the effect between Milnepan and death within 7 days. All of the results for multiple imputation showed that Milnepan could have protective effect, approximately reduce 12-18% odds. In the complete-case analyses, both logistic regression and propensity score weighting (PSW) demonstrated significant reductions in mortality (OR 0.88, 95% CI 0.79-0.99, p =

0.040; OR 0.72, 95% CI 0.65-0.81, p < 0.001). Although inverse probability of treatment weighting (IPTW) in the complete-case analysis did not statistic significant, a similar protective trend was observed (OR 0.92, 95%CI 0.82-1.03, p = 0.163). Overall, treatment with Milnepan was associated with reduced 7-day mortality, with the complete-case analyses serving as sensitivity analyses to support these findings.

Discussion

Both results for survival analysis and the causal analyses consistently suggested that Milnepan is associated with a modest reduction in 7-day mortality, corresponding to an approximately 12-18% lower odds of death compared with no Milnepan. However, the protective effect on clinical practice needs to be supported by biological mechanisms.

There are several limitations on this study. First, the dataset was derived from a single hospital, which may reflect local patterns. Therefore, data from multiple hospitals and regions are needed to improve generalisability. Additionally, imbalances in baseline characteristics could affect the evaluation of treatment effects. For instance, 72.0% of the sample were non-smokers, which may underestimate the impact of smoking. The prevalence of diabetes was low overall, 10.4%, while it was higher in the control group, 14.8%, which might cause by a consideration when prescribing the medication. Second, 50.11% of patients are recorded as day 10 or right censor, which might unable to capture longer-term benefits or harms of Milnepan. Future studies should extend follow-up beyond day 10 to better characterize the intervention's full clinical treatment. Third, although missing values in BMI and smoking were handled using standard multiple imputation method, the validity of the imputed values still relies on the MAR assumption and correct model specification. Additionally, the imputation model may be misspecified if important predictors of missingness or mortality were unavailable in the dataset. Although we included all available baseline variables in the imputation model to maximize efficiency, unmeasured clinical factors such as disease severity and comorbidities could influence both missingness patterns and outcomes, leading to imprecision or residual bias in imputed values. Thus, this concerned would also effect causal analysis. In the sensitivity analysis, using complete-case data may introduce bias due to the reduced sample size, which decreased from 8,192 to 5,280, and potential selection bias. Patients excluded because of missing BMI or smoking data tended to be older, have less diabetes, and higher mortality, suggesting that certain clinical profiles were disproportionately removed. This could lead to bias if the pattern of missingness is related to both treatment and outcome. Indeed, the complete-case IPTW analysis yielded a slightly attenuated and less precise effect estimate (OR 0.92, 95%CI 0.82-1.03, $p = 0.163$), highlighting the importance of appropriately handling missing data and interpreting complete-case analyses as sensitivity checks rather than primary results. Using datasets from different regions or hospitals may help validate these findings and mitigate potential biases.

In conclusion, treatment with Milnepan was associated with a reduced risk of death within 7 days. However, these findings should be interpreted with caution due to

several limitations mention above. Future studies using multi-centre or regional datasets are warranted to confirm the protective effect of Milnepan and to ensure that the findings are applied to broader populations.

Conclusions

In this study, Milnepan treatment was associated with a lower 7-day mortality. While promising, these results should be interpreted cautiously due to limitations such as single-centre data and missing information. Future research in diverse populations is warranted to validate these findings and guide clinical practice.